8(a)(i)	$T = RC\ln\left(\frac{\varepsilon - V_s}{\varepsilon - V_b}\right)$ $= 18\times10^{3}\times33\times10^{-6}\ln\left(\frac{5.4-0.9}{5.4-4.5}\right)$ [C1] $= 0.956\text{s}$ $f = \frac{1}{T} = 1.046\,\text{s}^{-1}$ = 1.05 Hz [M1] Comment: Passage mentioned: steady heart rate is 60 beats per minute, the frequency of 1.046 s^{-1} is about 62 beats per min which is close to steady heart rate. Hence it is appropriate. [A1] *Candidate must compare current frequency with steady heart rate from passage to gain this mark.* Award [B1] for any one physiologically valid situation where a fixed pacing rate is inappropriate, with a brief explanation of why. Examples: **Physical activity**: cannot increase rate during exercise leading to insufficient cardiac output during exertion **Sleep**: healthy hearts slow to 40–50 bpm during deep sleep hence fixed rate may be unnecessarily high, consuming more battery **Emotional stress/anxiety**: heart rate naturally rises and fixed rate cannot respond **Illness/fever**: body may require higher heart rate and fixed rate cannot adapt **Individual variation**: 60 bpm may not be the optimal resting rate for every patient but there is no way to adjust without surgery to replace resistor **Competing with intrinsic rhythm**: if patient's own heart occasionally fires, fixed-rate pacemaker may deliver a pulse at the wrong moment in the cardiac cycle, potentially dangerous p.s. this is actually the most clinically serious one, known as "R-on-T phenomenon"	**C1** **M1** **A1** **B1**
8(a)(ii)	τ_charging = RC = $18\times10^{3}\times33\times10^{-6}$ = 0.594 s □ τ_discharge = R_L C = $500\times33\times10^{-6}$ = 0.0165 s [M1 for both calculations] □ τ_discharge is 36× smaller than τ_charging (or compare and say about 2% only) ; since discharge time << charging time << T, the assumption that discharge is almost instantaneous is well justified. [A1]	**M1** **A1**
8(b)(i)	The energy U stored in a capacitor of capacitance C charged to potential difference V is: $U = \frac{1}{2}CV^2$ Since the capacitor charges to V_b before each discharge and retains charge at Vs after each discharge, the energy ΔU delivered to the heart per pulse is: $\Delta U = \frac{1}{2}C\left(V_b^{\,2} - V_s^{\,2}\right)$ = ½ (33 x 10^{-6})($4.5^2 - 0.9^2$) = 3.21 x 10^{-4} J	**C1** **M1** for correct working + substitution A0
8(b)(ii)	Power = energy / time = energy x frequency	

	$P = 3.21 \times 10^{-4} \times 1.046$ $= 0.0003357$ $= 3.36 \times 10^{-4}$ W	M1 A1
8(b)(iii)	Each pulse is said to be almost "instantaneous" which means the power is delivered only during the discharging process [B1] and during the charging process, no energy is delivered to the heart tissue [B1]; hence the time-averaged power is much lower than the power during each pulse	B1 B1
8(c)(i)	Converts capacity to coulombs: Q = 1.2 × 3600 = 4320 C **[C1]** Energy = VQ = 5.4 × 4320 = 23328 J = 23.3 kJ (3sf) **[A1]**	C1 A1
8(c)(ii)	Lifespan = E_batt / P_avg = 23328 / (3.36×10^{-4}) = 6.94×10^{7} s = 2.2 years Very short lifespan, will require operation every 2.2 years to change battery. OR 2.2 years is an overestimate because the calculation assumes all stored battery energy is ultimately delivered to the heart and neglects losses/other circuit consumption.	M1 A1
8(d)(i)	Increase in C → increase in T calculated from (a)(i)/ time constant → longer charging time and longer period T → lower pacing frequency since f = 1/T. [1]	B1
8(d)(ii)	A patient with complete heart block is entirely dependent on the pacemaker for ventricular stimulation; if the pacing rate drops, their heart rate drops correspondingly. [B1] If the pacing rate falls below ~40 bpm (as stated in passage), the patient risks fainting / cardiac arrest. [B1] A slow gradual drift may go unnoticed without regular monitoring, person's health may deteriorate slowly, making it potentially life-threatening / dangerous. [B1] Award max 2 points for any 2 points.	B1 B1
	TOTAL FOR QUESTION 8	**19**